

Chapter 100

Filter Nodes

This chapter details the Filter nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

Contents

Create Bump Map [CBu].....	2270
Custom Filter Node [CFIt].....	2272
Erode Dilate Node [ErDI].....	2276
Filter Node [Fitr].....	2278
Rank Filter Node [RFIt].....	2281
The Common Controls.....	2283

Create Bump Map [CBu]



The Create Bump Map node

Create Bump Map Node Introduction

The Create Bump Map node converts a grayscale (height map) image into a bump map. Unlike the Bump Map node that turns an image into a 3D material, the Create Bump Map node creates bump vector data and provides the output as an RGB image so other image-processing operations can be applied.

Input

The Create Bump Map node includes two inputs: one for the main image and the other for an effect mask to limit the area where the bump map is created.

Input: The orange input takes the RGBA channels from an image to calculate the bump map.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the creation of the bump map to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

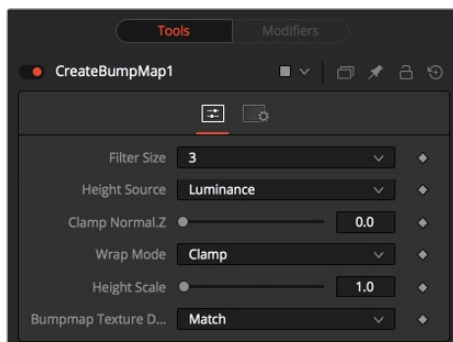
Basic Node Setup

The Create Bump Map node accepts a 2D grayscale image like a fast noise, which can then go through various 2D image-processing filters to create the bump map texture.



A Create Bump Map node produces a bump map as an RGB image for further image processing

Inspector



Create Bump Map controls

Controls Tab

The Controls tab contains all parameters for creating the bump map.

Filter Size

This menu sets the filter size for creating the bump map. You can set the filter size at 3 x 3 pixels or 5 x 5 pixels, thus determining the radius of the pixels sampled. The larger the size, the more time it takes to render.

Height Source

The Height Source menu selects the channel for extracting the grayscale information.

Clamp Normal.Z

This slider clips the lower values of the blue channel in the resulting bump texture.

Wrap Mode

This menu determines how the image wraps at the borders, so the filter produces a correct result when using seamless tiling textures.

Height Scale

The height scale menu modifies the contrast of the resulting values in the bump map. Increasing this value yields in a more visible bump map.

Bump Map Texture Depth

This menu matches or converts the resulting bump texture into the desired bit depth.

NOTE: The below definitions are provided to clarify some of the terminology used in the Create Bump Map node and other similar types of nodes.

Height Map: A grayscale image containing a height value per pixel.

Bump Map: An image containing normals stored in the RGB channels used for modifying the existing normals (usually given in tangent space).

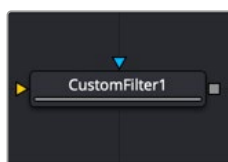
Normal Map: An image containing normals stored in the RGB channels used for replacing the existing normals (usually given in tangent or object space).

Common Controls

Settings Tab

The Settings tab controls are common to all Filter nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

Custom Filter Node [CFlt]



The Custom Filter node

Custom Filter Node Introduction

The Custom Filter node is used to apply custom convolution filters to images. A custom convolution filter can give a wide variety of image effects. For example, emboss, relief, sharpen, blurring, and edge detection are all convolution filters. There are many supplied custom filters in the Filters directory that can be loaded by right-clicking on the control header and selecting Settings > Load from the contextual menu.

The Custom filter uses an array (or grid) of either 3 x 3, 5 x 5, or 7 x 7 values. (Note: The array in the Inspector always shows a 7 x 7 grid; however, setting the Matrix Size to 3 x 3 uses only the center 9 cells.) The center of the array represents the current pixel, and entries nearby represent adjacent pixels. A value of 1 applies the full value of the pixel to the filter. A value of 0 ignores the pixel's value. A value greater than 1 multiplies the pixel's effect on the result. Negative values can also be entered, where the value of the pixel is subtracted from the average. Only integer values can be entered; 0.x is not valid.

Input

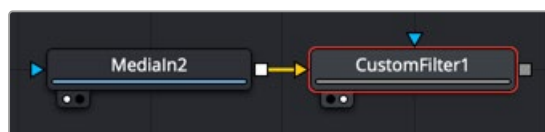
The Custom Filter node includes two inputs: one for the main image and the other for an effect mask to limit the area where the custom filter is applied.

Input: The orange input takes the RGBA channels from an image to calculate the custom filter.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the custom filter to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

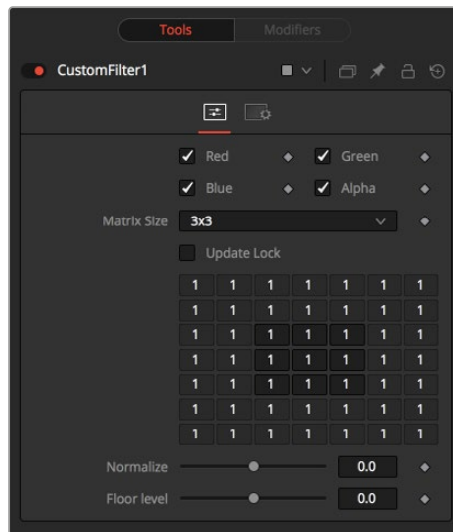
Basic Node Setup

The Custom Filter node can be inserted after an image, mask, or any node that needs a custom convolution filter applied.



A Custom Filter node placed after a MediaIn node in DaVinci Resolve

Inspector



Custom Filter controls

Controls Tab

The Controls tab is used to set the filter size and then use the filter matrix to enter convolution filter values.

Color Channels (RGBA)

The custom filter defaults to operating on R, G, B, and A channels. Selective channel editing is possible by enabling or disabling the checkboxes beside each channel.

This is not the same as the RGBA checkboxes found under the Common Controls. The node takes these controls into account before it processes. Deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, these controls under the Common Controls tab are applied after the node has processed.

Matrix Size

This menu is used to set the size of the filter at 3 x 3 pixels, 5 x 5 pixels, or 7 x 7 pixels, thus setting the radius of the pixels sampled. The larger the size, the more time it takes to render.

Update Lock

When this control is selected, Fusion does not render the filter. This is useful for setting up each value of the filter, and then turning off Update Lock and rendering the filter.

Filter Matrix

The Filter Matrix control is a 7 x 7 grid of text boxes where a number is entered to represent how much influence each pixel has on the overall convolution filter. The text box in the center represents the pixel that is processed by the filter. The text box to the left of the center represents the pixel to the immediate left, and so forth.

The default Matrix size is 3 x 3. Only the pixels immediately adjacent to the current pixel are analyzed. If a larger Matrix size is set, more of the text boxes in the grid are enabled for input.

Normalize

This controls the amount of filter normalization that is applied to the result. Zero gives a normalized image. Positive values brighten or raise the level of the filter result. Negative values darken or lower the level.

Floor Level

This adds or subtracts a minimum, or Floor Level, to the result of the filtered image. Zero does not add anything to the image. Positive values add to the filtered image, and negative values subtract from the image.

Examples

Original Image Example

For example, a filter with the values...

```
0 0 0
0 1 0
0 0 0
```

...has zero effect from its neighboring pixels, and the resulting image would be unchanged.



Original image

Softening Example

A slight softening effect would be...

```
1 1 1
1 1 1
1 1 1
```

...where the neighboring pixels are averaged with the center.



Before and after averaging neighboring pixels to soften the image

Emboss Example

The example below subtracts five times the value from the top left and adds five times the value from the lower right.

```
-5 0 0
0 1 0
0 0 5
```

If parts of the processed image are very smooth in color, the neighboring values are very similar.

In parts of the image where the pixels are different (e.g., an edge), the results are different and tend to highlight or emboss edges in the image.



A Custom Filter adding and subtracting neighboring pixels to create an embossed image

Exposure Example

Using the values...

```
1 1 1
1 1 1
1 1 1
```

...and adjusting Normalize to a positive value makes the image brighter or glow, simulating film overexposure.



The Custom Filter Normalize slider used to change exposure

Relief Example

Using the values...

```
-1 0 0
0 0 0
0 0 1
```

... and adjusting Floor Level to a positive value creates a Relief filter.



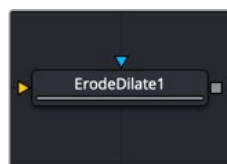
Custom Filter Floor Level slider used to create a relief

Common Controls

Settings Tab

The Settings tab controls are common to all Filter nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

Erode Dilate Node [ErDI]



The Erode Dilate node

Erode Dilate Node Introduction

The Erode Dilate node contracts or expands the image, depending on whether the Amount slider is set to a negative or positive value.

Inputs

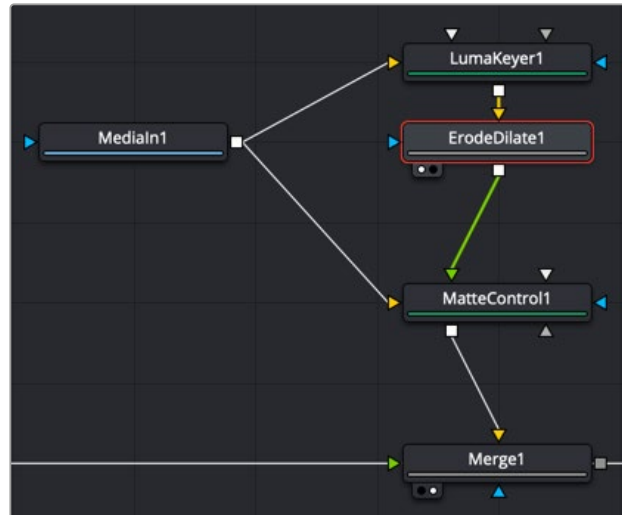
The Erode Dilate node includes two inputs: one for the main image and the other for an effect mask to limit the area where the erode or dilate is applied.

Input: The orange input takes the RGBA channels from an image to calculate the custom filter.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the erode or dilate to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

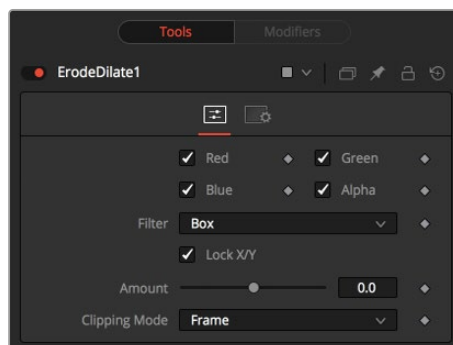
Basic Node Setup

The Erode Dilate node is commonly used to contract or expand mattes. Below, a Luma Keyer is connected to the Erode Dilate and passes the modified key to a Matte Control where it is embedded into the image.



An Erode Dilate node placed after a Luma Keyer to operate on a Matte Control

Inspector



Erode Dilate controls

Controls Tab

The Controls tab includes the main Amount slider that determines whether you are performing an erode by entering a negative value or a dilate by entering a positive value.

Color Channels (RGBA)

The Erode Dilate node defaults to operating on R, G, B, and A channels. Selective channel editing is possible by enabling or disabling the checkboxes beside each channel.

This is not the same as the RGBA checkboxes found under the Common Controls. The node takes these controls into account before it processes. Deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, the channel controls under the Common Controls tab are applied after the node has processed.

Lock X/Y

The Lock X/Y checkbox is used to separate the Amount slider into amount X and amount Y, allowing a different value for the effect on each axis.

Amount

A negative value for Amount causes the image to erode. Eroding simulates the effect of an underexposed frame, shrinking the image by growing darker areas of the image so that they eat away at brighter regions.

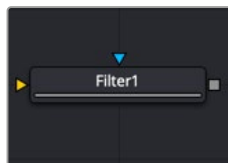
A positive value for Amount causes the image to dilate, similar to the effect of overexposing a camera. Regions of high luminance and brightness grow, eating away at the darker regions of the image. Both techniques eradicate fine detail in the image and tend to posterize fine gradients.

The Amount slider scale is based on the input image width. An amount value of 1 = image width. So, if you want to erode or dilate by exactly 1 pixel on an HD image, you would enter 1/1920, or 0.00052083.

Common Controls**Settings Tab**

The Settings tab controls are common to all Filter nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

Filter Node [Fltr]



The Filter node

Filter Node Introduction

The Filter node contains several standard convolution filters, easily selectable from a list. This node enables a variety of effects, from radically changing the look of an image to adding subtle randomly-generated film grain. The Sobel and Laplacian settings are often used for edge detection.

Inputs

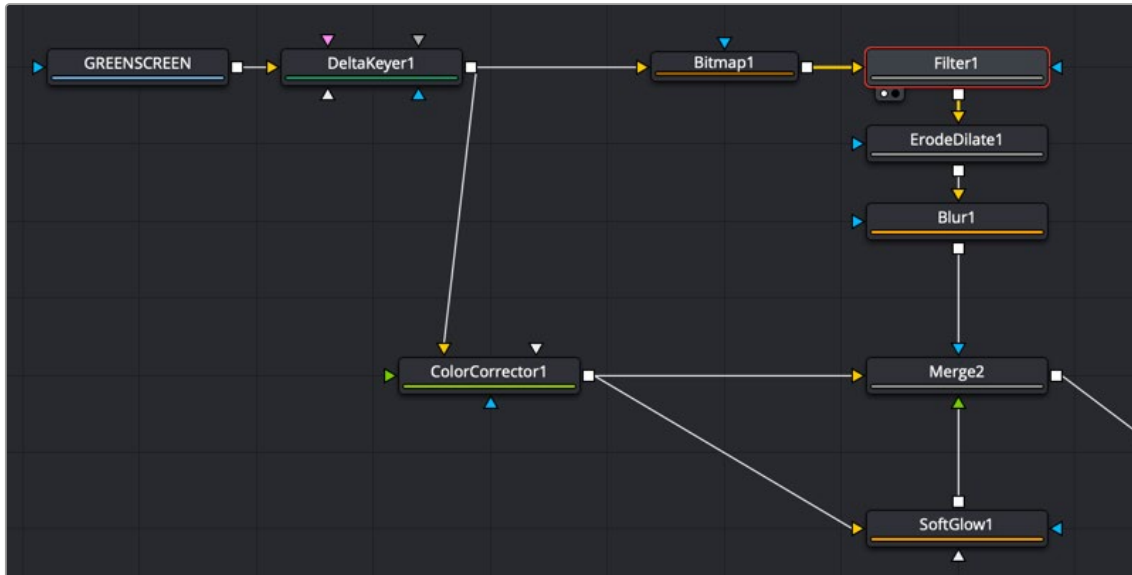
The Filter node includes two inputs: one for the main image and the other for an effect mask to limit the area where the filter is applied.

Input: The orange input is used for the primary 2D image that gets the filter applied.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the filter to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

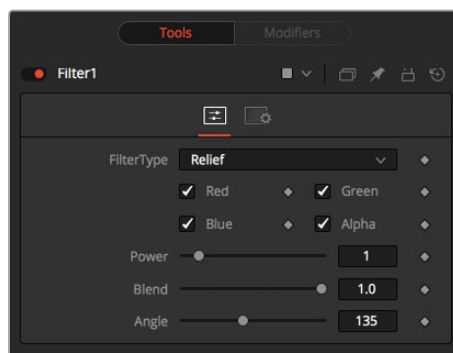
Basic Node Setup

The Filter node can be inserted after an image, mask, or any node that needs a filter applied. Below, it is used to create an edge matte, which is then used to mask the soft glow around the keyed foreground.



A Filter node using the Sobel setting to extract an edge matte

Inspector



Filter controls

Controls Tab

The Controls tab is used to set the filter type, the channels the filter is applied to, and the amount it blends with the original image.

Filter Type

The Filter Type menu provides a selection of filter types described below.

- **Relief:** This appears to press the image into metal, such as an image on a coin. The image appears to be bumped and overlaid on gray.
- **Emboss Over:** Embosses the image over the top of itself, with adjustable highlight and shadow height and direction.

- **Noise:** Uniformly adds noise to images. This is often useful for 3D computer-generated images that need to be composited with live action, as it reduces the squeaky-clean look that is inherent in rendered images. The frame number acts as the random generator seed. Therefore, the effect is different on each frame and is repeatable.
- **Defocus:** This filter type blurs the image.
- **Sobel:** Sobel is an advanced edge detection filter. Used in conjunction with a Glow filter, it creates impressive neon light effects from live-action or 3D-rendered images.
- **Laplacian:** Laplacian is a very sensitive edge detection filter that produces a finer edge than the Sobel filter.
- **Grain:** Adds noise to images similar to the grain of film (mostly in the midrange). This is useful for 3D computer-generated images that need to be composited with live action as it reduces the squeaky-clean look that is inherent in rendered images. The frame number acts as the random generator seed. Therefore, the effect is different on each frame and is repeatable.

Color Channels (RGBA)

The Filter node defaults to operating on R, G, B, and A channels. Selective channel editing is possible by enabling or disabling the checkboxes beside each channel.

Power

Values range from 1 to 10. Power proportionately increases the amount by which the selected filter affects the image. This does not apply to the Sobel or Laplacian filter type.

Angle

This control has a range from 0 to 315 degrees and changes the effect in increments of 45 degrees. This applies only to the Relief and Emboss filters.

Median

Depending on which Filter Type is selected, the Median control may appear. It varies the Median filter's effect. A value of 0.5 produces the true median result, as it finds the middle values. A value of 0.0 finds the minimums, and 1.0 finds the maximums. This applies to the Median setting only.

Seed

This control is visible only when applying the Grain or Noise filter types. The Seed slider can be used to ensure that the random elements of the effect are seeded with a consistent value. The randomizer always produces the same result, given the same seed value.

Animated

This control is visible only when applying the Grain or Noise filter types. Select the checkbox to cause the noise or grain to change from frame to frame. To produce static noise, deselect this checkbox.

Common Controls

Settings Tab

The Settings tab controls are common to all Filter nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

Rank Filter Node [RFlt]



The Rank Filter node

Rank Filter Node Introduction

The Rank Filter examines nearby pixels, sorts the pixels by value, and then replaces the color of the examined pixels with the color of the pixel with the selected rank.

Inputs

The Rank Filter node includes two inputs: one for the main image and the other for an effect mask to limit the area where the filter is applied.

Input: The orange input is used for the primary 2D image that gets the Rank filter applied.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the rank filter to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

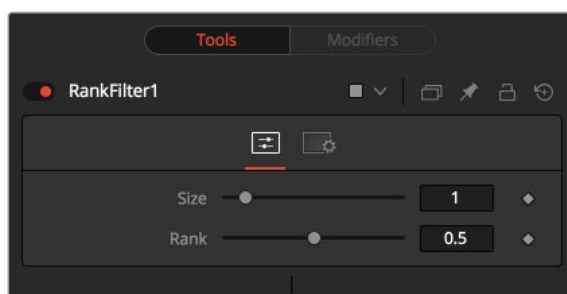
Basic Node Setup

The Rank Filter node can be placed anywhere in a node tree to apply an effect to an image.



A Rank Filter node placed after a Median node in DaVinci Resolve

Inspector



Rank Filter controls

Controls Tab

The Controls tab is used to set the size and rank value of the filter.

Size

This control determines the size in pixels of the area sampled by the filter. A value of 1 samples 1 pixel in each direction, adjacent to the center pixel. This produces a total of 9 pixels, including the center sampled pixel. Larger values sample from a larger area.

Low Size settings are excellent for removing salt and pepper style noise, while larger Size settings produce an effect similar to watercolor paintings.

Rank

The Rank slider determines which value from the sampled pixels is chosen. A value of 0 is the lowest value (darkest pixel), and 1 is the highest value (brightest pixel).

Example

Below is a before and after example of a Rank filter with Size set to 7 and a Rank of 0.7 to create a watercolor effect.

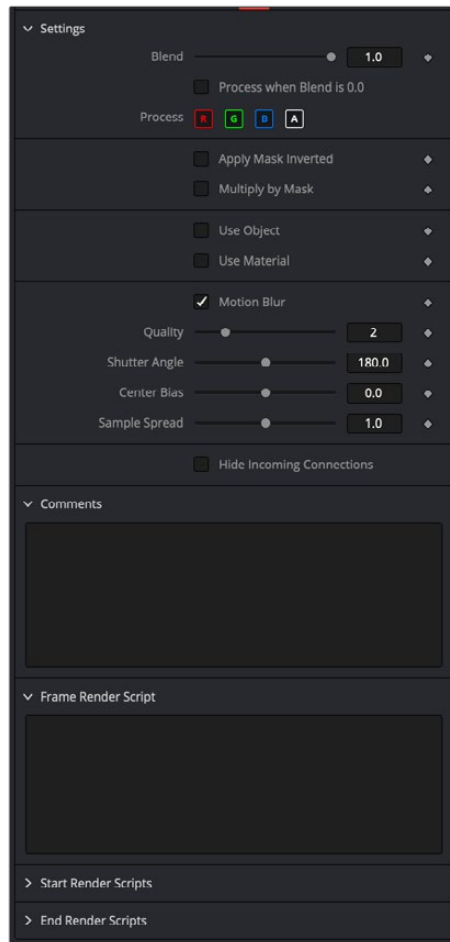


Before and after of a Rank Filter producing a watercolor-style effect

The Common Controls

Filter nodes share a number of identical controls in the Inspector. This section describes controls that are common among Filter nodes.

Inspector



Common Filter settings Inspector

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Filter category. The Settings controls are even found on third-party filter-type plugin tools. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options, which are also covered here.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.